

Machine Learning Model Performance Report

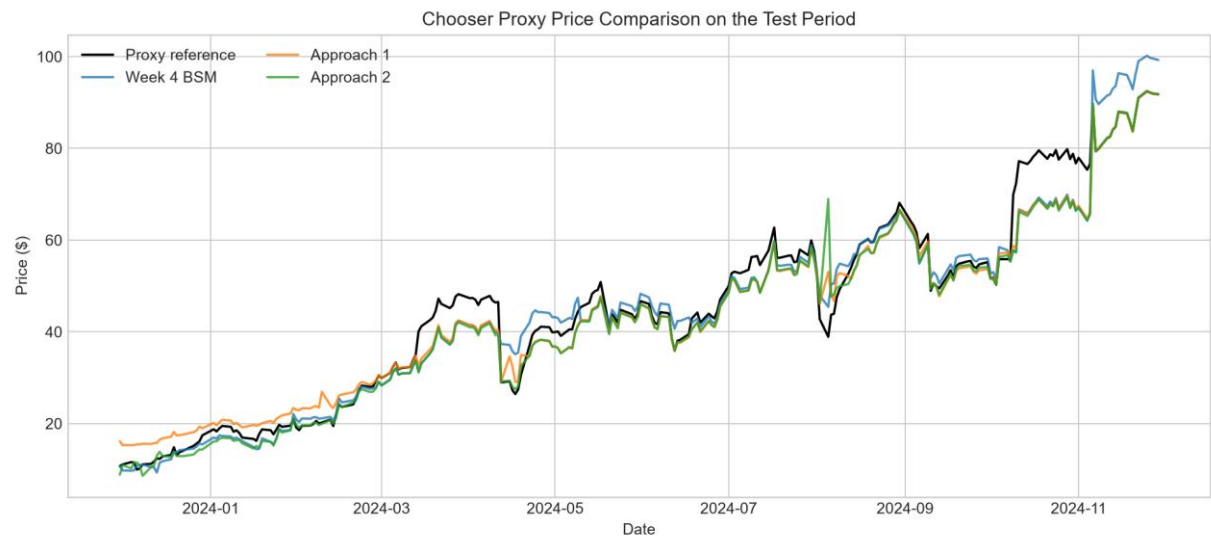
The Week 6 comparison selects Random Forest volatility forecasting followed by BSM. It reduced test RMSE by 11.30% relative to the Week 4 BSM baseline while preserving the same 252 test dates and proxy target.

Selection Method

Expanding-window CV uses a 20-day gap. Approach 1 retains validation-based selection. Following the live negative-price failure, Approach 2 now learns a bounded normalized time-value fraction with constrained moneyiness tails. Training CV chooses regularization and distance scaling. The 2023-11-30 to 2024-11-29 test is reused retrospectively for this revision, not treated as a pristine holdout.

Approach	MAE	RMSE	R ²	Mean error
RF vol plus BSM	3.1384	4.3203	0.9575	-1.2781
Trailing 20D BSM	3.5698	4.8706	0.9460	-0.5299
Bounded direct price	2.8501	4.5404	0.9530	-2.3689

Figure 1 Historical test price comparison



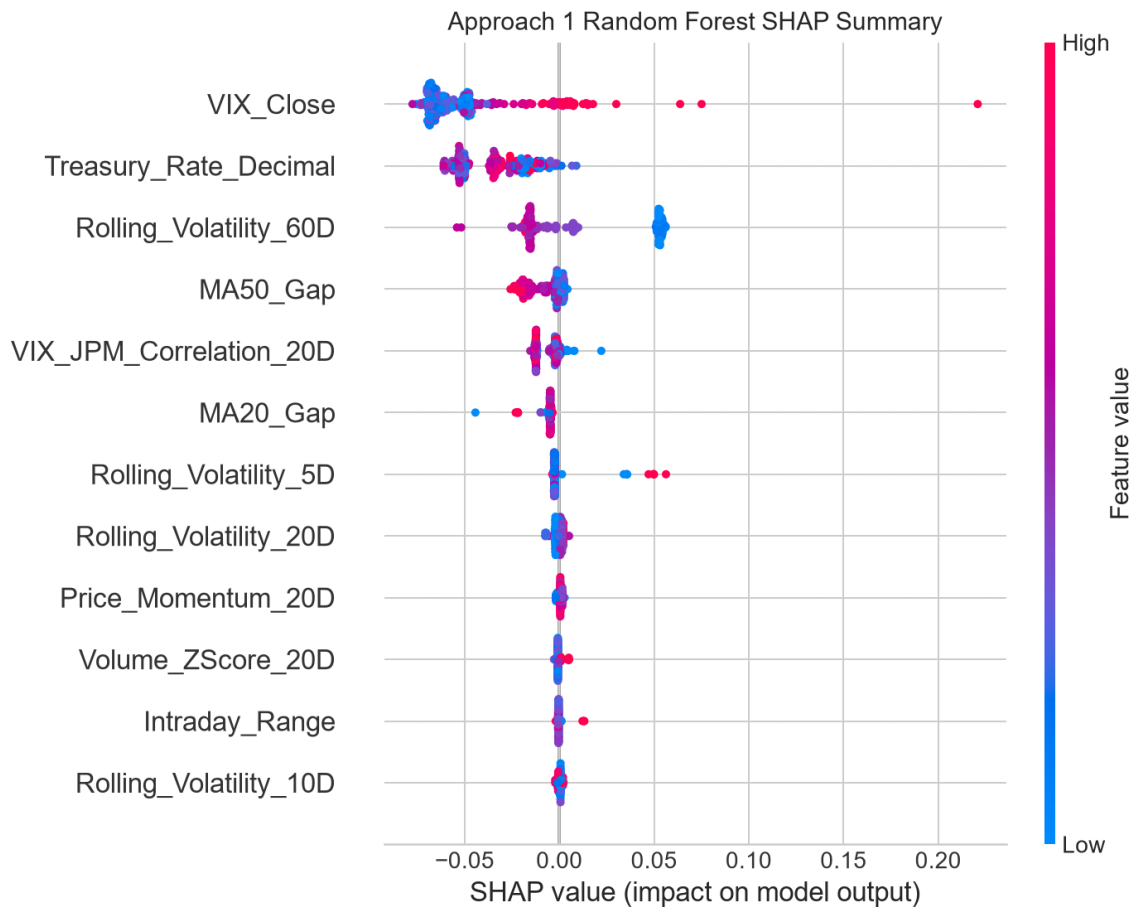
Volatility Forecast

Forecast	MAE	RMSE
Persistence	0.085584	0.120650
Selected Random Forest	0.072424	0.094069

Model Interpretation

Tree SHAP explains the selected Random Forest volatility predictor. Permutation SHAP explains the complete bounded direct-price function in dollar units, using training-only background data. VIX-related shares are 40.80% and 23.29%, respectively. Shares measure fitted behavior, not causal effects.

Figure 2 Approach 1 SHAP summary



Serialized Model Deliverables

Both selected estimators are delivered as native Python pickle files, as required by the Week 6 rubric. Each pickle was reloaded and checked against all 252 historical test rows; the maximum absolute difference from the saved Week 6 inference outputs was below $1e-12$. Joblib copies remain available for the Week 8 application.

Artifact	Selected model	Delivery role
best_volatility_model.pkl	Random Forest volatility forecaster	Required pickle deliverable for Approach 1
best_direct_pricing_model.pkl	Bounded time-value pricer	Required pickle deliverable for Approach 2
Two .joblib copies	Same fitted estimators	Runtime compatibility for the Week 8 tool

Decision and Limits

- Use Random Forest volatility plus BSM as the selected ML route in Week 8.
- Retain the bounded direct-price route as a comparison model. Its validation RMSE is 7.3983 versus the legacy 5.4837; its retrospective test RMSE improves from 49.9891 to 4.5404. A separate post-freeze 2025–2026 cached-data diagnostic gives RMSE 3.7972 on 404 proxy targets. It uses later downloaded adjusted history, not point-in-time vintages or real trades. Pointwise bounds do not prove full-surface arbitrage freedom.
- Display the BSM baseline beside every ML estimate.
- Warn when current inputs exceed the fitted-sample range.
- Describe every score as proxy-target performance.